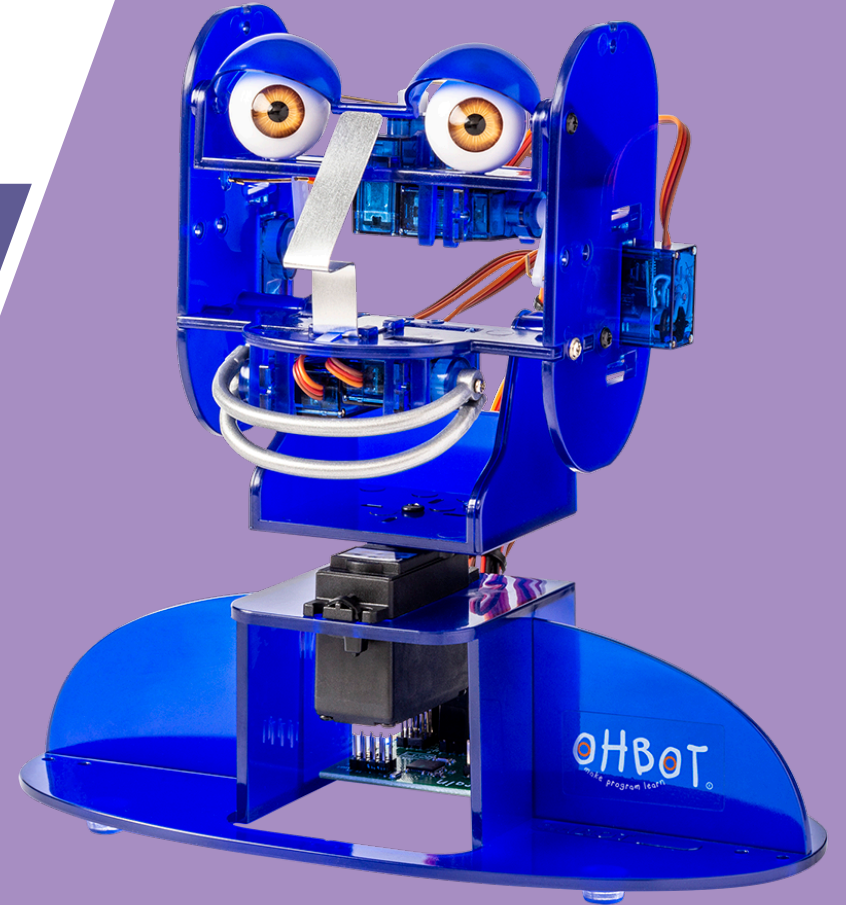


Ubiquitous Systems Lab

A Behaviors Engine for the OhBot

Hassan Khalid Butt
Sabina Mahmudova
Jouhaina Salsabil El Euch

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<https://learn.ohbot.co.uk/store/p22/OhbotAssembled.html>

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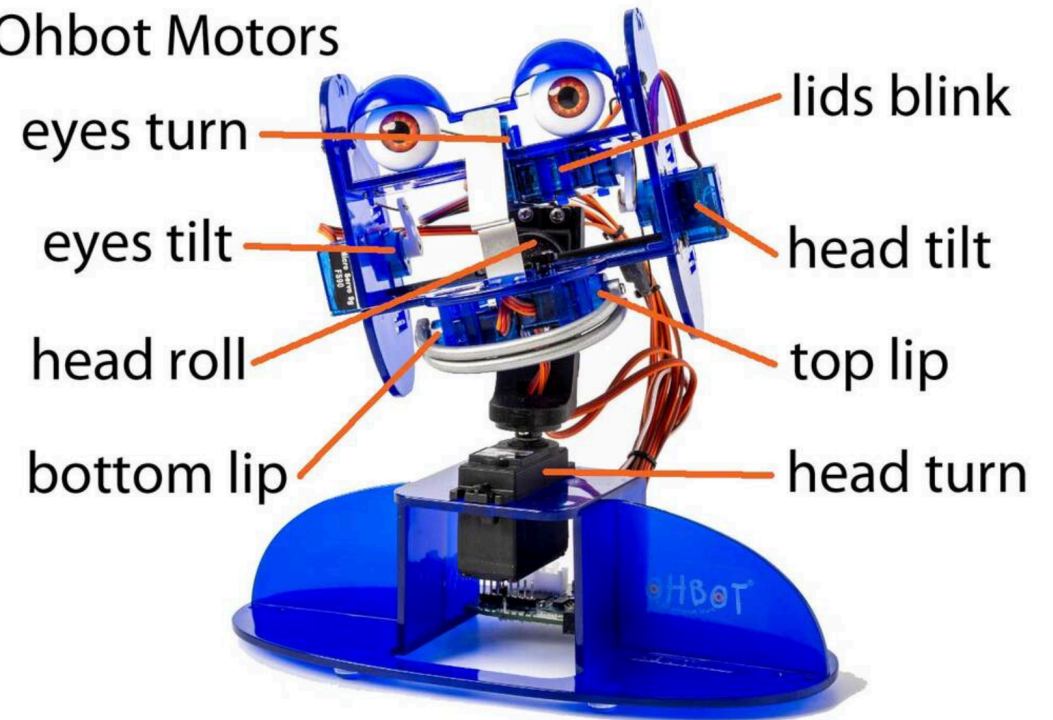
Q&A

Introduction

Introduction

- OhBot = humanoid social robot
- 8 Motors
- Speak using text-to-speech
- Display emotions through motions
- Technologies:
 - OhBot Python Library
 - Google Gemini API
 - SpeechRecognition

Ohbot Motors

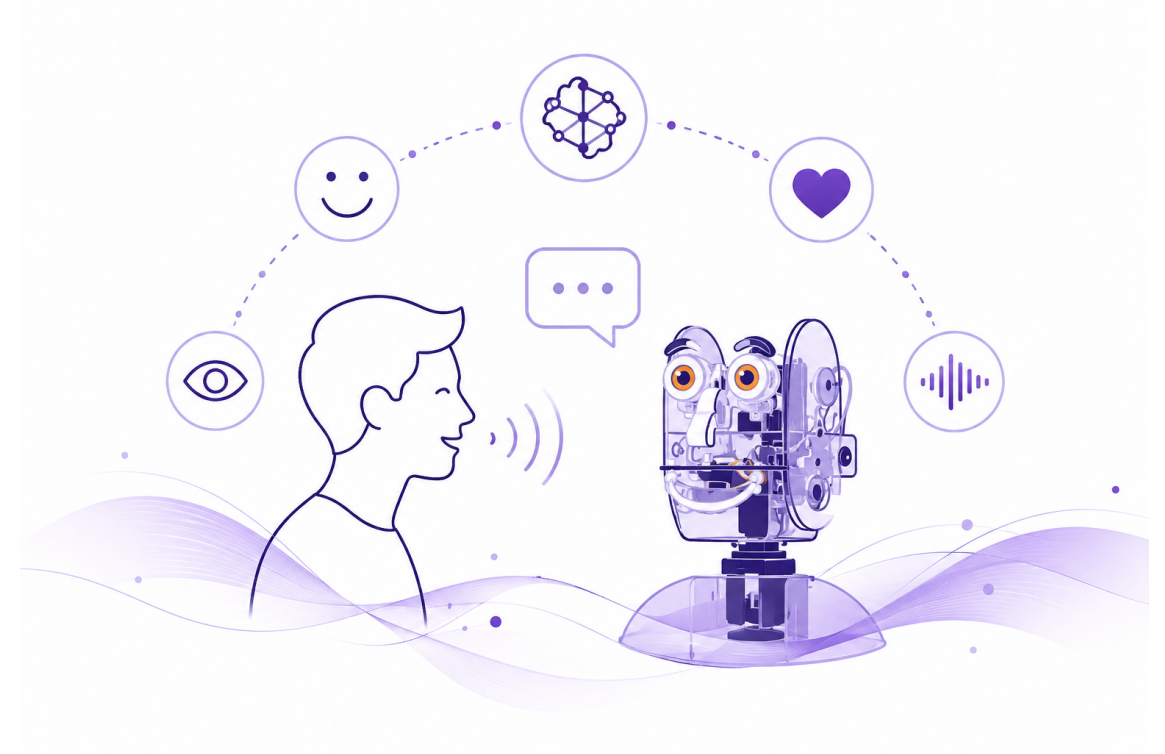


<https://learn.ohbot.co.uk/blockprog.html>

Goal of the project

Goal of the project

- React dynamically while speaking
 - Show emotions through movements
 - Behave less mechanically
 - Create a more engaging interaction with users
-
- Main focus:
 - Facial expressions
 - Idle behaviors
 - AI-generated responses
 - Multithreading for simultaneous actions

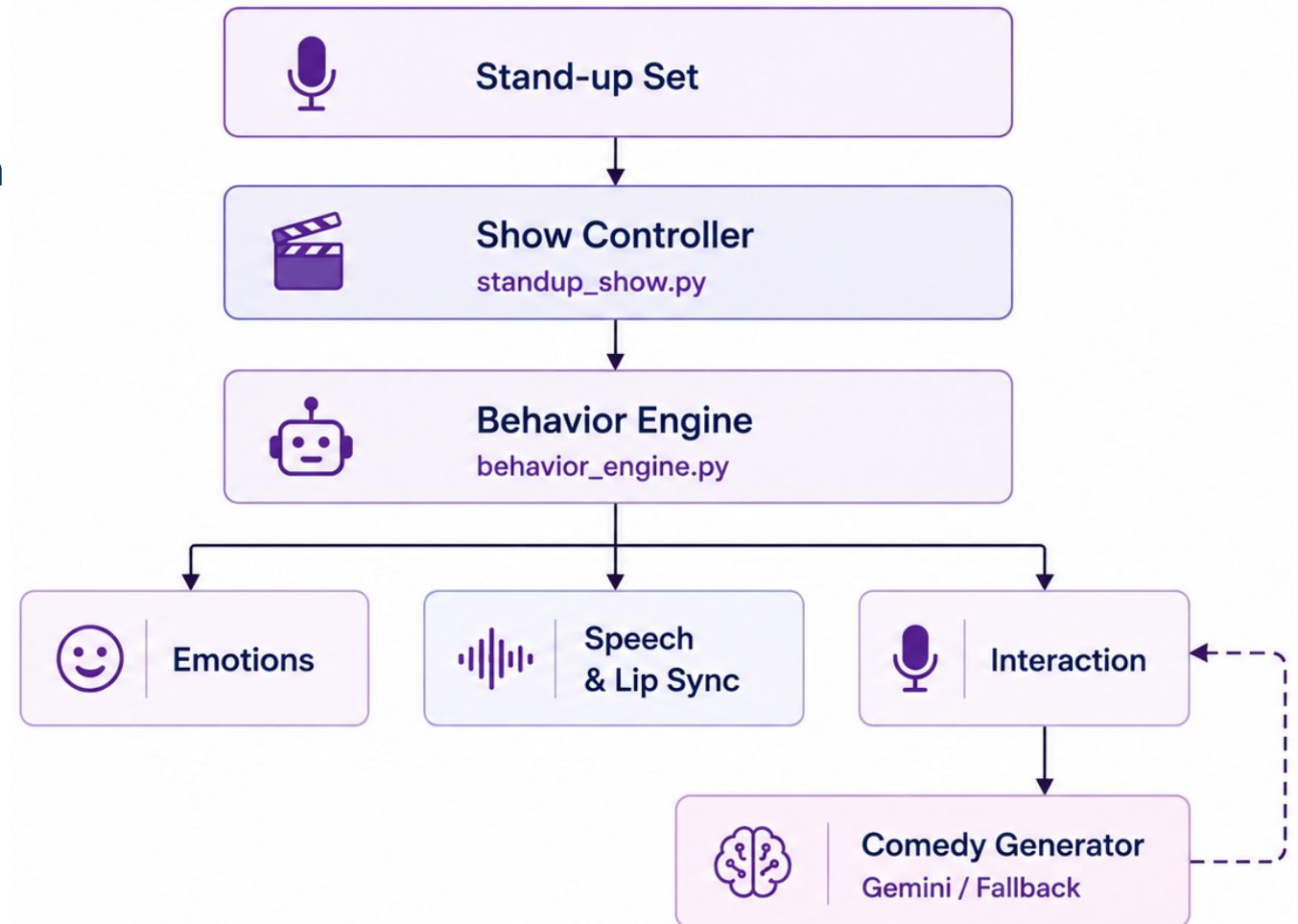


➡ From a speaking robot to an expressive social companion

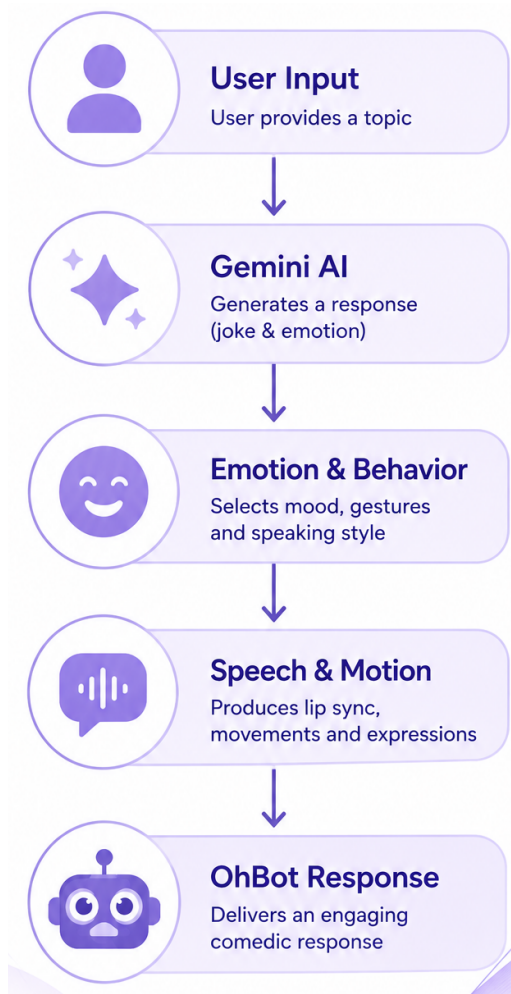
Final Implementation

Final Implementation

- Emotion System:
 - happy, sad, angry, surprised, neutral
 - new: disappointed and side-eye expression
- Motion & Realism
 - Natural blinking with random intervals
 - Idle head and eye movements
 - Micro-movements during speech
 - Multi-threaded behavior management
 - Made emotions more realistic
- Speech & Lip Sync
 - Heuristic lip synchronization
 - Different mouth shapes (visemes)
 - Natural speech timing with pauses



Final Implementation



- AI
 - Gemini API integrated
 - AI-generated stand-up comedy routines
 - Fallback jokes if Gemini is unavailable
 - Sentence-level emotional tagging
- Interaction
 - Text & Voice input using microphone
 - Users can provide comedy topics
 - Interactive comedy session implemented
 - Voice recognition from 1 word to whole sentence
 - added better jokes

Demo

Challenges & Solutions

Challenges & Solutions

Challenge	Solution
Speech recognition unreliable	Calibration, retries, keyboard fallback
Emotion disappeared too quickly	Mood held throughout the sentence
Gemini unavailable	Local fallback jokes
Parallel movements	Threaded behavior management
Lip sync not phoneme-accurate	Lightweight viseme heuristic

The innovation is not only joke generation, but the coordination of language, timing, emotion and physical behavior.

Thank you for your attention!

Q&A

Sources

- <https://learn.ohbot.co.uk>
- <https://ai.google.dev/gemini-api/docs?hl=de>
- Images generated with ChatGPT (OpenAI): <https://chatgpt.com>